3/23/2022

Wednesday, March 23, 2022 9:18 AM

M168_Notes_7.4

7.4 Use of Counting Techniques in Probability

Let S be a uniform sample space and E be any event in S. Then

$$P(E) = \frac{\text{number of outcomes in event E}}{\text{number of outcomes in the Sample Space}} = \frac{n(E)}{n(S)}$$

Example

Suppose we have a jar with 8 blue and 6 green marbles. What is the probability that in a sample of 2, both will be blue?

$P(E) = \frac{4\times 7}{7\times 13} = \frac{4}{13}$

$\rightarrow C(14,2) = \frac{14!}{12!\times 2!} = 7\times 13 = n(S) = \frac{14!}{12!\times 2!} = \frac{14\times 13}{2} = 7\times 13$

$\rightarrow C(8,2) = \frac{8!}{6!\times 2!} = n(E) = 4\times 7$

$\frac{C(8,2)}{C(14,2)}$

What is the probability there is at least one blue marble?

$n(E) = \boxed{C(8,1)\times C(6,1)} + C(8,2)$

$= 8\times 6 + 4\times 7 = 48 + 28 = 76$

$P(E) = \frac{76}{91}$

8B
6G

$C(8,1)\times C(6,1) = 8\times 6 = 48$

Find the probability distribution table for the number of blue marbles in the sample of 2 marbles:

Events	Probability
0 Blue	$C(6,2)/C(14,2) = \frac{15}{91}$
1 Blue	$\frac{48}{91}$
2 Blue	$\frac{28}{91}$

$15 + 48 + 28 \stackrel{?}{=} 91$

Example

A stack of 100 copies has 3 defective papers. What is the probability that in a sample of 10 there will be no defective papers?

$\rightarrow n(S) = C(100, 10) = \dfrac{100!}{90! \times 10!}$

$\rightarrow n(E) = C(97, 10) = \dfrac{97!}{87! \times 10!}$

$P(E) = \dfrac{n(E)}{n(S)} = \dfrac{97! \times 90! \times \cancel{10!}}{100! \times \cancel{87! \times 10!}}$

$= \dfrac{C(97, 10)}{C(100, 10)} = \dfrac{90 \cdot 89 \cdot 88}{100 \cdot 99 \cdot 98}$

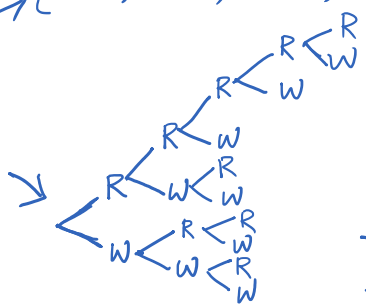
$= \dfrac{178}{245}$

Example

A student takes a true/false test with 5 questions by guessing (choose answer at random). Write a probability distribution table for the number of correct answers.

$\rightarrow \{RRR, RRW, RWR, RWW, WRR, WRW, WWR, WWW\}$

$n(S) = 2^5 = 32$



Event	Probability
0 R	$\frac{1}{32}$
1 R	$\frac{5}{32}$
2 R	$\frac{10}{32}$
3 R	$\frac{10}{32}$
4 R	$\frac{5}{32}$
5 R	$\frac{1}{32}$

$_\ _\ _\ _\ _$ $C(5, 1) = 5$

$C(5, 2) = 10$